A conformally calibrated linear-fractional robust model for food-indexed cash transfers and price monitoring under partial information

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Abstract

Food-indexed cash transfers protect recipients only if payments track local prices, yet planners usually observe prices through national or zonal averages, a few state extremes and delayed surveys. This paper develops a robust allocation model for a fixed transfer budget under such partial information. We show that the protection a transfer rule delivers factors exactly into budget adequacy and allocative efficiency, and that over any polytope of admissible prices the allocation with the best worst-case efficiency has a closed form: each state is paid in proportion to its maximal need share, obtained from a linear-fractional program solved as a linear program. For grouped price polytopes and for boxes on state needs the shares are computed exactly without a solver, and a water-filling rule trades nominal performance against the guarantee. Calibrating the uncertainty set by conformal prediction turns the guarantee into a statement with explicit coverage. We then choose which states to monitor so that the guarantee rises fastest. Using 125,598 state–item–month food prices for Nigeria from January 2017 to August 2023, weighted by the food budgets of poor households, a flat transfer delivered on average 82% of attainable protection and 70% in the worst month. Indexing to zonal averages, the information published since September 2023, did worse than the flat rule in 34 of 54 months. Over 2021 to 2023 a guarantee-constrained rule realised 91% of attainable protection, against 78% for the flat rule, while certifying 64% in the worst case, against 40%. Certificates held in every month and adaptive calibration covered true needs in 87% of months at an 80% target. Monitoring five states without delay raised the guarantee from 0.72 to 0.79, but a one-month reporting delay removed most of the gain. From September 2023 to May 2026, when no state food prices are published, the guarantee fell from 0.88 to 0.67 and the flat rule's certificate from 0.70 to 0.39. The approach gives statistical agencies and social-protection programmes a transparent way to value local price data.

Keywords: Robust optimization ; Linear-fractional programming ; Conformal prediction ; Cash transfers ; Food price monitoring ; Nigeria

1 Introduction

In mid-2023 Nigeria removed its petrol subsidy and unified its exchange-rate windows [6, 28]. Food prices rose quickly, but not uniformly. In the state-level price data of the National Bureau of Statistics (NBS), the cost of a fixed 43-item food basket rose by very different amounts across the 36 states and the Federal Capital Territory. Between January 2019 and mid-2021, the growth of a poor household's food basket cost in the most exposed state was 2.0 times that in the least exposed state (section 7). Governments and humanitarian agencies responded with cash transfers. The federal programme launched in October 2023 targeted 15 million households with a uniform N75,000 paid in three instalments, financed in part by an $800 million World Bank operation [20]. Such transfers are typically paid at one national rate.

A uniform payment protects recipients unequally when local prices diverge. Recipients in states where food became most expensive can buy the smallest share of what the transfer was meant to buy. The obvious remedy is to index payments to local prices. That runs into an information problem. Planners rarely observe current local prices. They see national or zonal averages, the names and prices of the most and least expensive states, and survey results that arrive weeks or months late. In the NBS releases we examined, the last full state-level food price table is for August 2023, and none exists for June 2023. Since September 2023 only national and zonal averages and the highest- and lowest-priced state per item have been released [21]. Some compiled datasets fill later months by copying zonal averages to every state in the zone, which removes exactly the variation that matters here.

This paper asks how a fixed transfer budget should be allocated across states under such partial information, how much protection that allocation can guarantee, and which prices are worth collecting to raise the guarantee. We answer these questions with a model that is simple enough to be computed in a spreadsheet and rigorous enough to carry a statistical guarantee.

1.1 Research questions

(Q1) How should a fixed transfer budget be allocated across states so that every recipient retains the same share of a food entitlement, when local prices are observed only through aggregates, extremes and delayed samples?

(Q2) How much protection can such an allocation guarantee, and how can that guarantee be made statistically valid rather than assumed?

(Q3) Which state prices should a monitoring system collect, and how quickly, to raise the guarantee most per observation?

1.2 Contributions

1. *An equal-protection allocation model with an exact decomposition.* The protection a rule delivers equals budget adequacy times allocative efficiency (proposition 1). The decomposition separates the question of how much money is available from the question of where it goes.

2. *A closed-form robust solution and fast exact computation.* Over any polytope of admissible prices, the allocation with the best worst-case efficiency pays each state in proportion to its maximal need share. Its guarantee is the reciprocal of the recipient-weighted sum of those shares (theorem 1). Each share is a linear-fractional program, solved exactly as a linear program by the Charnes–Cooper transformation [7] (proposition 2). For polytopes built from group averages the shares are found by sorting and a Dinkelbach iteration (proposition 3), and for boxes on needs they have a closed form (proposition 4). A water-filling rule spans the frontier between nominal and worst-case performance (proposition 5).

3. *Statistical validity.* The price polytope is calibrated by conformal prediction [26, 29], with an adaptive variant for non-exchangeable time series [16]. The guarantee then holds with stated probability (proposition 6).

4. *Monitoring design.* We choose which prices to observe to maximise the expected guarantee, prove that the objective is monotone (proposition 7), and trace an information–protection curve over the number of monitored states and the reporting delay. Decision-focused monitoring outperforms accuracy-focused monitoring, and timeliness matters more than coverage.

5. *Evidence from Nigeria.* We backtest the model on 125,598 genuine state–item–month prices from 2017 to 2023, and apply it to September 2023 to May 2026, when only aggregates and state extremes are published.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 sets out the model. Section 4 derives the robust allocation and its guarantee. Section 5 calibrates the uncertainty set. Section 6 treats monitoring design. Section 7 presents the Nigerian case study. Section 8 discusses policy implications and section 9 concludes.

2 Literature review

The paper draws on five strands of work: the measurement of spatial price differences, the design of cash and food assistance, robust and data-driven optimisation, conformal prediction for decisions, and the design of monitoring systems.

2.1 Spatial price variation and welfare measurement

Prices differ across regions of large countries, and welfare comparisons that ignore those differences misrank households and places [10, 11]. Food prices matter most for the poor, whose budgets are dominated by food, and in many low-income countries consumer price indices are compiled from a few urban markets. Dietrich et al. [12] show for seven West African countries that replacing national price indices with local market prices changes who is classified as food poor, with inclusion and exclusion errors of about 5% and 2%, and they call for systematic collection of local prices to support transfers indexed to prices. Andrée [2] goes further on the data side: machine-learning imputation recovers market-level food price inflation from partial and intermittent surveys in 25 fragile countries. This literature measures how much local price information matters for poverty statistics. We take the next step and ask how a budget should be allocated when that information is incomplete, and how much the missing information costs in guaranteed protection.

2.2 Allocation of cash transfers and food assistance

Most quantitative work on cash transfers concerns targeting: deciding who receives a transfer when poverty is observed with error. Aiken et al. [1] show that phone data and machine learning can improve targeting of humanitarian aid. Zheng et al. [32] study how much to give each beneficiary when targeting errors are known in distribution, compare predictive, stochastic and robust allocation rules, and show that a predictive rule is optimal when targeting errors are well calibrated. Humanitarian practice adjusts transfer values to the cost of a minimum expenditure basket, recommends separate values by area when local prices diverge from national trends, and triggers revisions when cumulative price changes exceed a threshold [25]. In operations research, food assistance has been studied as a supply chain under uncertain procurement prices [9] and as an equitable distribution problem for food banks [8, 14]. Our setting differs from all of these. Recipients are known; the uncertainty is in the prices they face. The decision is a regional allocation of a fixed budget, the objective is equal protection, and the information available to the planner is itself a design variable.

2.3 Robust and data-driven optimisation

Robust optimisation protects a decision against every realisation in an uncertainty set [3], and the conservatism of the result depends on the size and shape of that set [4]. Data-driven variants build the set from samples with statistical guarantees [5]. Robust and bi-level models are increasingly used in socio-economic planning, for example for location and pricing in agricultural

markets [31]. Our problem has a ratio objective. Linear-fractional programmes reduce to linear programmes by the transformation of Charnes and Cooper [7], and parametric methods such as Dinkelbach [13] solve them by a sequence of linear problems; we use both. Predict-then-optimise methods train forecasts for the decisions they feed [15]. Our guarantee-constrained rule is a simple, transparent way to combine a point forecast with a worst-case certificate, and the realised–certified frontier makes the trade-off explicit.

2.4 Conformal prediction for decisions

Conformal prediction gives finite-sample, distribution-free coverage for prediction sets under exchangeability [26, 29], and adaptive conformal inference controls the long-run miss rate under distribution shift [16]. Several recent papers use conformal sets as uncertainty sets for robust optimisation: Johnstone and Cox [17] calibrate ellipsoidal sets, Sun et al. [27] calibrate the uncertainty of a fixed predictor for robust contextual linear programmes, Patel et al. [24] build conformal regions from generative models, and Yeh et al. [30] learn the uncertainty set end to end. These papers treat generic objectives and continuous predictors. We exploit the structure of an allocation problem: with a box on needs the robust decision and its guarantee are available in closed form, a monthly simultaneous score over 37 states gives a coverage statement for the guarantee itself, and horizon-indexed calibration handles information that becomes stale.

2.5 Value of information and design of price monitoring

Humanitarian and statistical agencies maintain price-monitoring networks, and the question of where to collect data is usually settled by coverage and logistics. In the sensor-placement literature, locations are chosen to maximise an information criterion, typically by greedy algorithms with guarantees for submodular objectives [18, 19, 23]. We choose monitored states by the guarantee of the downstream allocation instead of a statistical criterion, compare this decision-focused design with an accuracy-focused design, and add reporting delay as a design dimension.

2.6 Distinction of our work

Table 1 compares the closest papers. To our knowledge no existing model allocates a fixed transfer budget across regions under partial information about local prices, gives a closed-form robust allocation with a calibrated guarantee, and uses that guarantee to value and design price monitoring.

3 Problem description and model

3.1 Setting and notation

Let S be the set of states ($|S| = 37$, including the Federal Capital Territory) and J a reference basket of food items. In month t, $p_{sjt} > 0$ is the average retail price of item j in state s, and $p_t \in \mathbb{R}_+^{S\times J}$ stacks these prices. A programme with $N_s > 0$ recipients in state s promises each recipient a food entitlement worth T_0 in a base month 0, where the base basket has quantities $q_j \geq 0$ and costs $C_{s0} = \sum_j q_j p_{sj0}$ in state s. Keeping that entitlement intact in month t requires

$$D_s(p_t) \;=\; T_0\,\frac{\sum_j q_j\, p_{sjt}}{C_{s0}} \;=\; a_s^\top p_t, \qquad a_{sj} = \frac{T_0\, q_j}{C_{s0}} \text{ on the block of state } s, \tag{1}$$

so need is linear in prices. We drop the subscript t when no confusion arises. The planner has a budget $B > 0$ and chooses payments $T = (T_s)_{s\in S}$ in the feasible set

$$\mathcal{T}(B) = \Big\{T \in \mathbb{R}_+^S :\; \textstyle\sum_s N_s T_s = B\Big\}. \tag{2}$$

Table 1: Positioning of this paper. "Guarantee": a stated bound on decision quality with explicit coverage. "Info. design": the model chooses what data to collect.

Study	Decision	Uncertain quantity	Uncertainty model	Guarantee	Info. design
Zheng et al. [32]	transfer per beneficiary	household welfare	distribution, robust set	no	no
Aiken et al. [1]	who receives aid	household poverty	ML prediction	no	no
Dawood and van Vuuren [8]	food-bank redistribution	food insecurity	ML index	no	no
de Moor et al. [9]	food aid procurement	procurement prices	robust, adaptive	no	no
Zhang and Liu [31]	location and pricing	demand, prices	robust bi-level	no	no
Dietrich et al. [12]	(measurement)	local food prices	observed vs. national CPI	no	no
Andrée [2]	(measurement)	missing market prices	ML imputation	no	no
Johnstone and Cox [17], Patel et al. [24], Yeh et al. [30]	generic	generic	conformal sets	yes	no
This paper	regional transfer allocation	local food prices and needs	conformal polytope and need box	yes	yes

3.2 Protection, adequacy and efficiency

Definition 1 (Protection)**.** The protection delivered by T at prices p is the share of the entitlement guaranteed to every recipient,

$$\rho(T,p) = \min_{s\in S} \frac{T_s}{D_s(p)}. \tag{3}$$

Proposition 1 (Full information)**.** *For known prices p, $\max_{T\in\mathcal{T}(B)} \rho(T,p) = \rho^*(p) = B/\sum_k N_k D_k(p)$, attained uniquely by $T_s = \rho^*(p)\, D_s(p)$. Consequently, for every $T \in \mathcal{T}(B)$,*

$$\rho(T,p) = \rho^*(p) \times E(T,p), \qquad E(T,p) = \frac{\rho(T,p)}{\rho^*(p)} \in (0,1], \tag{4}$$

*where $\rho^*(p)$ depends only on the budget (*adequacy*) and $E(T,p)$ is invariant to rescaling the budget (*allocative efficiency*).*

Proof. Let $T \in \mathcal{T}(B)$ and $\rho = \rho(T,p)$. Then $T_s \geq \rho D_s(p)$ for all s, so $B = \sum_s N_s T_s \geq \rho \sum_s N_s D_s(p)$, that is $\rho \leq \rho^*(p)$. The allocation $T_s = \rho^*(p) D_s(p)$ is feasible and attains the bound. If T attains the bound, every inequality $T_s \geq \rho^* D_s$ must hold with equality because $N_s > 0$, which gives uniqueness. Finally, if T is replaced by cT and B by cB for $c > 0$, both $\rho(T,p)$ and $\rho^*(p)$ scale by c, so E is unchanged. □

The decomposition is the policy device of the paper. A budget that is too small lowers adequacy for everyone; a budget in the wrong places lowers efficiency. For the uniform rule $T_s \equiv B/\sum_k N_k$,

$$E_{\text{flat}}(p) = \frac{\sum_k N_k D_k(p) / \sum_k N_k}{\max_s D_s(p)}, \tag{5}$$

the recipient-weighted mean need divided by the largest need.

3.3 Partial information as a price polytope

At decision time the planner holds an information set $\mathcal{I}_t$, which we translate into a set $\mathcal{U}_t \subset \mathbb{R}_+^{S\times J}$ of price vectors consistent with it. All information used here gives linear constraints:

(U1) *monitored prices*: $p_{sjt} = o_{sjt}$ for (s, j) in a monitored set M;

(U2) *published national averages* A_{jt}: $(1-\varepsilon)A_{jt} \le |S|^{-1}\sum_s p_{sjt} \le (1+\varepsilon)A_{jt}$;

(U3) *published zonal averages*: the same constraint for each geopolitical zone;

(U4) *published extremes* H_{jt} and L_{jt} attained in named states h and l: $p_{hjt} = H_{jt}$, $p_{ljt} = L_{jt}$ and $L_{jt} \le p_{sjt} \le H_{jt}$ for all s;

(U5) *calibrated relative-price bands*: $\ell_{sjt}A_{jt} \le p_{sjt} \le u_{sjt}A_{jt}$ (section 5);

(U6) *delayed observations* from d months earlier: $g^{\ell}_{sjt}\,p_{sj,t-d} \le p_{sjt} \le g^{u}_{sjt}\,p_{sj,t-d}$.

Assumption 1. $\mathcal{U}_t$ is a nonempty bounded polytope, $\mathcal{U}_t = \{p \ge 0 :\ \underline{p} \le p \le \overline{p},\ Rp \le r\}$, and $D_s(p) > 0$ for all s and $p \in \mathcal{U}_t$.

The audit in section 7 calibrates ε: published national averages equal the unweighted mean of state prices in 78% of item–months, and the tolerance is set from the distribution of the remaining gaps.

4 Robust allocation and its guarantee

Define the *need share* of state s and its worst case over $\mathcal{U}$:

$$\pi_s(p) = \frac{D_s(p)}{\sum_k N_k D_k(p)}, \qquad \pi_s^{\max}(\mathcal{U}) = \max_{p\in\mathcal{U}} \pi_s(p). \tag{6}$$

Note that $\sum_s N_s\pi_s(p) = 1$ for every p, and the maximum in (6) exists because π_s is continuous on the compact set $\mathcal{U}$.

Theorem 1 (Closed-form robust allocation)**.** *Under assumption 1,*

$$\max_{T\in\mathcal{T}(B)}\ \min_{p\in\mathcal{U}}\ E(T,p) \;=\; G(\mathcal{U}) \;=\; \frac{1}{\sum_s N_s\,\pi_s^{\max}(\mathcal{U})}, \tag{7}$$

and the maximum is attained uniquely by

$$T_s^{\mathrm{rob}} = B\,\frac{\pi_s^{\max}(\mathcal{U})}{\sum_k N_k\,\pi_k^{\max}(\mathcal{U})}, \qquad s \in S. \tag{8}$$

Moreover $0 < G(\mathcal{U}) \le 1$*, with* $G(\mathcal{U}) = 1$ *when* $\mathcal{U}$ *is a single point.*

Proof. Fix $T \in \mathcal{T}(B)$. By proposition 1,

$$E(T,p) = \frac{\min_s T_s/D_s(p)}{B/\sum_k N_k D_k(p)} = \min_{s\in S}\frac{T_s}{B\,\pi_s(p)}.$$

Minimising over $p \in \mathcal{U}$ and exchanging the two minimisations gives

$$\min_{p\in\mathcal{U}} E(T,p) = \min_{s\in S}\ \min_{p\in\mathcal{U}}\frac{T_s}{B\,\pi_s(p)} = \min_{s\in S}\frac{T_s}{B\,\pi_s^{\max}(\mathcal{U})} =: \theta(T).$$

Since $T_s \ge \theta(T)\,B\,\pi_s^{\max}$ for all s, the budget gives $B = \sum_s N_s T_s \ge \theta(T)\,B\sum_s N_s\pi_s^{\max}$, so $\theta(T) \le G(\mathcal{U})$. The allocation (8) is feasible and has $T_s/(B\pi_s^{\max}) = G(\mathcal{U})$ for every s, so it attains the bound. If $\theta(T) = G(\mathcal{U})$, every inequality $T_s \ge G\,B\,\pi_s^{\max}$ holds with equality because $N_s > 0$, which proves uniqueness. For any $p_0 \in \mathcal{U}$, $\sum_s N_s\pi_s^{\max} \ge \sum_s N_s\pi_s(p_0) = 1$, so $G \le 1$; if $\mathcal{U} = \{p_0\}$ the inequality is an equality. □

Algorithm 1 Robust allocation in month t

Require: information set $\mathcal{I}_t$; calibrated bands; recipients N; budget B

1: Build $\mathcal{U}_t = \{p : \underline{p} \le p \le \overline{p},\ Rp \le r\}$ from (U1)–(U6)
2: **for** $s \in S$ **do**
3: Solve LP (9); store $\pi_s^{\max}$ and $p^{(s)} = y^*/\tau^*$
4: **end for**
5: $G_t \leftarrow 1/\sum_s N_s \pi_s^{\max}$; $T_s \leftarrow B\,\pi_s^{\max} G_t$ for all s
6: **return** T, G_t and the adversarial price vectors $\{p^{(s)}\}$

Corollary 1 (Monotonicity)**.** *If $\mathcal{U}' \subseteq \mathcal{U}$ then $\pi_s^{\max}(\mathcal{U}') \le \pi_s^{\max}(\mathcal{U})$ for all s and hence $G(\mathcal{U}') \ge G(\mathcal{U})$. Additional information never lowers the guarantee.*

Remark 1. The guarantee in (7) is a competitive ratio: whatever prices materialise in $\mathcal{U}$, T^{rob} delivers at least a fraction $G(\mathcal{U})$ of the protection that a planner with full information could deliver with the same budget. The adversarial prices differ by state: state s is protected against the price vector that maximises its own need share.

Proposition 2 (Computation by linear programming)**.** *Under assumption 1, for each s,*

$$\pi_s^{\max}(\mathcal{U}) = \max_{y \ge 0,\ \tau \ge 0} \Big\{ a_s^\top y :\ Ry \le r\tau,\ \underline{p}\,\tau \le y \le \overline{p}\,\tau,\ \textstyle\sum_k N_k\, a_k^\top y = 1 \Big\}, \tag{9}$$

and an optimal p in (6) *is recovered as y^*/τ^*.*

Proof. Write $d(p) = \sum_k N_k a_k^\top p > 0$ on $\mathcal{U}$. For $p \in \mathcal{U}$, the pair $(y, \tau) = (p/d(p), 1/d(p))$ is feasible in (9) with objective $\pi_s(p)$, so the optimal value of (9) is at least $\pi_s^{\max}$. Conversely, let (y, τ) be feasible. If $\tau = 0$, then $Ry \le 0$ and $0 \le y \le 0$, so $y = 0$, which contradicts $\sum_k N_k a_k^\top y = 1$. Hence $\tau > 0$, $p = y/\tau \in \mathcal{U}$, $d(p) = 1/\tau$, and $a_s^\top y = \pi_s(p) \le \pi_s^{\max}$. The two values are equal and the maximiser maps as stated. □

Computing the robust allocation therefore requires $|S| = 37$ linear programs per month and information scenario (algorithm 1).

4.1 Two tractable uncertainty sets

Proposition 2 needs a general LP solver and, with 1,554 price cells, a well-scaled formulation. Two structures that arise in practice are solved faster and more accurately.

Grouped price polytopes. Suppose the cells are partitioned into groups $g \in \mathcal{G}$ (for example, the states of one zone for one item) and

$$\mathcal{U} = \Big\{ p :\ \underline{p} \le p \le \overline{p},\ \ \underline{r}_g \le \textstyle\sum_{c \in g} p_c \le \overline{r}_g\ \ (g \in \mathcal{G}) \Big\}. \tag{10}$$

This covers (U1), (U3), (U4) and (U5) together, and (U2) and (U5) together; monitored cells and published extremes become cells with $\underline{p}_c = \overline{p}_c$. Write $b = \sum_k N_k a_k$, so that $\pi_s(p) = a_s^\top p / b^\top p$.

Proposition 3 (Exact computation for grouped polytopes)**.** *Let $\mathcal{U}$ be given by* (10) *and nonempty, and let $F_s(\lambda) = \max_{p \in \mathcal{U}} (a_s - \lambda b)^\top p$. Then:*

(a) $\pi_s^{\max}(\mathcal{U})$ is the unique root of F_s, which is convex and strictly decreasing;

(b) for fixed λ, $F_s(\lambda)$ separates into one continuous knapsack per group. With $c = a_s - \lambda b$, each group is solved by starting from $p = \underline{p}$, raising cells in decreasing order of c_c to their upper bounds until the group sum reaches $\underline{r}_g$ (even if $c_c < 0$), and then continuing only while $c_c > 0$ and the sum is below $\overline{r}_g$;

(c) the Dinkelbach iteration $\lambda_{k+1} = a_s^\top p_k / b^\top p_k$, with p_k a maximiser of $F_s(\lambda_k)$, increases strictly until it reaches $\pi_s^{\max}(\mathcal{U})$, which it does after finitely many steps.

Proof. (a) F_s is a maximum of affine functions of λ, hence convex, and each has slope $-b^\top p < 0$ by assumption 1, so F_s is strictly decreasing. $F_s(\lambda) \geq 0$ if and only if $a_s^\top p \geq \lambda b^\top p$ for some $p \in \mathcal{U}$, that is, $\lambda \leq \pi_s^{\max}$. (b) The constraints of (10) couple only cells in the same group, so the maximum is the sum of the group maxima. Within a group the problem is a fractional knapsack with unit weights, a lower and an upper capacity: any feasible point can be moved, without lowering $c^\top p$, to the point that fills cells in decreasing order of c_c, by transferring mass from a cell with smaller c to one with larger c; mass beyond $\underline{r}_g$ is added only where $c_c > 0$. (c) This is Dinkelbach's method. Since $F_s(\lambda_k) = a_s^\top p_k - \lambda_k b^\top p_k$, $\lambda_{k+1} - \lambda_k = F_s(\lambda_k)/b^\top p_k > 0$ while $\lambda_k < \pi_s^{\max}$. The maximiser in (b) is a vertex of $\mathcal{U}$, which has finitely many, and the ratios strictly increase, so no vertex repeats. □

Each evaluation of F_s costs one sort per group, and all states can be processed in one vectorised pass. On the Nigerian panel the iteration stops after about three evaluations, runs about 600 times faster than solving (9) with HiGHS, and returns maximisers whose constraint violations are of order 10^{-12}, while the LP solver stops 0.06% short of the optimum on this badly scaled problem.

Need boxes. Theorem 1 uses prices only through the need vector, so it holds for any compact set $\mathcal{D}$ of positive need vectors in place of $\{D(p) : p \in \mathcal{U}\}$. When uncertainty is modelled directly on the $|S|$ needs,

$$\mathcal{D} = \{D : \; L_s \leq D_s \leq U_s, \; s \in S\}, \qquad 0 < L \leq U, \tag{11}$$

the worst-case shares have a closed form.

Proposition 4 (Need box)**.** *For* (11)*,*

$$\pi_s^{\max} = \frac{U_s}{N_s U_s + \sum_{k \neq s} N_k L_k}, \qquad G = \Big(\sum_s N_s \pi_s^{\max} \Big)^{-1}.$$

Proof. $\pi_s = D_s / (N_s D_s + \sum_{k \neq s} N_k D_k)$ increases in D_s and decreases in each D_k, $k \neq s$, and the box constrains the coordinates independently. □

The need box replaces 1,554 simultaneous price bands by 37 need bands. As section 7.4 shows, this matters: a simultaneous band over many cells must be wide, and the price polytope then yields a weak guarantee.

4.2 Trading nominal performance for the guarantee

T^{rob} maximises the worst case and ignores the most likely prices. A planner with a point forecast $\hat{D}$ may prefer to pay in proportion to $\hat{D}$ and accept a weaker guarantee. The two aims combine in one problem. For $\kappa \in [0, 1]$,

$$\max_{T \in \mathcal{T}(B)} \; \min_{s \in S} \frac{T_s}{\hat{D}_s} \quad \text{s.t.} \quad \theta(T) = \min_s \frac{T_s}{B\, \pi_s^{\max}} \;\geq\; \kappa\, G. \tag{12}$$

Proposition 5 (Guarantee-constrained allocation)**.** *Let $f_s = \kappa\, G\, B\, \pi_s^{\max}$. The solution of* (12) *is*

$$T_s^\kappa = \max\{f_s, \; \vartheta\, \hat{D}_s\}, \tag{13}$$

where ϑ is the smallest value with $\sum_s N_s \max\{f_s, \vartheta \hat{D}_s\} = B$. It interpolates between proportional indexation to $\hat{D}$ ($\kappa = 0$) and T^{rob} ($\kappa = 1$), and T^κ is certified at worst-case efficiency at least κG.

Proof. The constraint in (12) is $T_s \geq f_s$ for all s, and $\sum_s N_s f_s = \kappa B \leq B$, so the problem is feasible. Let $h(\vartheta) = \sum_s N_s \max\{f_s, \vartheta \hat{D}_s\}$, which is continuous and nondecreasing with $h(0) = \kappa B$ and $h(\vartheta) \to \infty$. For any feasible T with nominal value $\phi = \min_s T_s/\hat{D}_s$, $T_s \geq \max\{f_s, \phi \hat{D}_s\}$, so $B \geq h(\phi)$ and $\phi \leq \vartheta^{\max} = \sup\{\vartheta : h(\vartheta) = B\}$. If $\kappa < 1$, h is strictly increasing at its root, the root is unique, and some state has $f_s < \vartheta \hat{D}_s$ (otherwise $h(\vartheta) = \kappa B < B$), so T^κ attains $\phi = \vartheta$. If $\kappa = 1$ the floors exhaust the budget and the only feasible point is T^{rob}. The certificate holds because $T^\kappa_s \geq f_s$. □

The water-filling rule is easy to explain: every state receives at least its insurance floor f_s, and the rest of the budget tops up states in proportion to the forecast. Varying κ traces a frontier between realised and certified efficiency (fig. 2).

5 Conformal calibration of the price polytope

The guarantee of theorem 1 is only as credible as $\mathcal{U}_t$. Constraints (U1)–(U4) restate published or observed facts. The bands (U5) and (U6) are statistical, and we calibrate them so that they contain the true prices with stated probability.

Let $x_{sjt} = \log(p_{sjt}/A_{jt})$ be the relative price. On a training sample, fit a centre $\hat{m}_{sjt}$ and scale $\hat{\sigma}_{sjt} > 0$ (section 7 compares several models). On a disjoint calibration sample of months $t = 1, \ldots, n$, compute the simultaneous score

$$R_t = \max_{s \in S,\, j \in J} \frac{|x_{sjt} - \hat{m}_{sjt}|}{\hat{\sigma}_{sjt}}, \tag{14}$$

and let $\hat{q}$ be the $\lceil (n+1)(1-\alpha) \rceil$-th smallest of $R_1, \ldots, R_n$. The bands in (U5) are $\ell_{sjt} = \exp(\hat{m}_{sjt} - \hat{q}\hat{\sigma}_{sjt})$ and $u_{sjt} = \exp(\hat{m}_{sjt} + \hat{q}\hat{\sigma}_{sjt})$.

Proposition 6 (Coverage of the guarantee)**.** *Suppose $R_1, \ldots, R_n, R_{n+1}$ are exchangeable, and that the constraints (U1)–(U4) hold for the true prices of month $n+1$. Then* $\Pr\{p_{n+1} \in \mathcal{U}_{n+1}\} \geq 1 - \alpha$*, and therefore*

$$\Pr\left\{E(T^{\text{rob}}_{n+1}, p_{n+1}) \geq G(\mathcal{U}_{n+1})\right\} \;\geq\; 1 - \alpha.$$

Proof. By exchangeability the rank of R_{n+1} among $R_1, \ldots, R_{n+1}$ is uniform (with ties broken at random), so $\Pr\{R_{n+1} \leq \hat{q}\} \geq 1 - \alpha$ [29]. On this event every relative price of month $n+1$ lies in its band, so p_{n+1} satisfies (U5); (U1)–(U4) hold by assumption, so $p_{n+1} \in \mathcal{U}_{n+1}$. By theorem 1, $E(T^{\text{rob}}, p) \geq G(\mathcal{U}_{n+1})$ for every $p \in \mathcal{U}_{n+1}$. □

Monthly prices are not exchangeable, especially across a policy shock. We therefore also use adaptive conformal inference [16], which updates the level after each month by $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, where err_t indicates a miss. It guarantees that the long-run average miss rate converges to α without exchangeability. We report empirical coverage for both versions.

Need bands. The same construction applies to the need box (11). With a plug-in forecast $\hat{D}_{st}$, the residuals $r_{st} = \log(D_{st}/\hat{D}_{st})$ receive a rolling centre $\hat{c}_{st}$ and scale $\hat{\sigma}_{st}$, the score is $R_t = \max_s |r_{st} - \hat{c}_{st}|/\hat{\sigma}_{st}$, and $L_s, U_s = \hat{D}_{st} e^{\hat{c}_{st}} \exp(\mp \hat{q}\, \hat{\sigma}_{st})$. Proposition 6 then holds with $\mathcal{D}_{n+1}$ in place of $\mathcal{U}_{n+1}$.

Stale information. When state data stop, the forecast for month t must be built from the last observed month $t - h$. We use the zonal growth forecast

$$\hat{D}^{(h)}_{st} = D_{s,t-h} \, \frac{D^Z_{st}}{D^Z_{s,t-h}}, \tag{15}$$

Algorithm 2 Greedy monitoring design

Require: candidate states $\mathcal{C}$; budget K; delay d; calibrated boxes for training months $\mathcal{T}_{\text{tr}}$

$M \leftarrow \emptyset$
for $k = 1, \ldots, K$ **do**
  for $u \in \mathcal{C} \setminus M$ **do**
    $\Phi(M \cup \{u\}) \leftarrow |\mathcal{T}_{\text{tr}}|^{-1} \sum_t G(\mathcal{D}_t(M \cup \{u\}))$ ▷ proposition 4
  end for
  $M \leftarrow M \cup \{\arg\max_u \Phi(M \cup \{u\})\}$
end for
return the nested sequence of monitored sets

where D^Z_{st} is the need of state s evaluated at its zone's average prices. For each horizon h we compute residuals of (15) on all historical months, a per-state scale $\sigma_{s,h}$ and a simultaneous conformal quantile q_h. Because sample sizes shrink with h, the raw widths $q_h\sigma_{s,h}$ need not grow with h; we use $w_{s,h} = \max_{h' \le h} q_{h'}\sigma_{s,h'}$. The bands are then nested in h, and by corollary 1 the guarantee can only fall as information ages.

6 Monitoring design

Observing prices shrinks the polytope. Let $\mathcal{C}$ be a set of candidate monitoring units (whole states, or state–item pairs) with costs $c_u > 0$, and let $\mathcal{U}_t(M)$ be the polytope when units $M \subseteq \mathcal{C}$ are observed. Because monitoring must be decided before prices are seen, we choose M on training months $\mathcal{T}_{\text{tr}}$:

$$\max_{M \subseteq \mathcal{C}} \ \Phi(M) = \frac{1}{|\mathcal{T}_{\text{tr}}|} \sum_{t \in \mathcal{T}_{\text{tr}}} G\big(\mathcal{U}_t(M)\big) \quad \text{s.t.} \quad \sum_{u \in M} c_u \le K. \tag{16}$$

Proposition 7 (Monotonicity of the monitoring objective)**.** *If the observed prices are the true prices, then $M \subseteq M'$ implies $\Phi(M) \le \Phi(M')$.*

Proof. Observing more units adds equality constraints satisfied by the true prices, so $\mathcal{U}_t(M') \subseteq \mathcal{U}_t(M)$ and both are nonempty. Apply corollary 1 month by month and average. □

We do not establish submodularity of Φ, so the $1 - 1/e$ guarantee of the greedy algorithm [23] is not claimed. We solve (16) by greedy forward selection (algorithm 2); lazy evaluation [19] reduces the work further. We measure the gap to the exact optimum by enumeration for $K \le 3$. As a benchmark we run the same procedure with an accuracy objective, the recipient-weighted squared error of the plug-in need forecast on training months, to test whether decision-focused monitoring outperforms accuracy-focused monitoring.

A monitored state reports with delay $d \in \{0, 1, 3\}$ months. With $d = 0$ its need is known exactly. With $d > 0$ its need is forecast by (15) with $h = d$ and receives its own calibrated band, which is narrower than the band of an unmonitored state but not zero.

7 Case study: Nigeria

7.1 Data

We use NBS Selected Food Price Watch data for 36 states and the Federal Capital Territory, covering 43 items monthly from January 2017 to August 2023 except June 2023 (125,598 state–item–month observations), together with published national and zonal averages and state extremes to May 2026 [21]. We verified every state value from April to August 2023 against the NBS state tables. We exclude 14,319 rows for June 2023 and September 2023 to April 2024 that

circulate in compiled datasets but repeat zonal averages for every state. We use the latest NBS vintage for 153 cells that NBS revised, and replace 82 transient spikes (a relative-price jump of more than one log point that reverses by at least half the next month) by interpolation of the relative price; results with the raw data are reported in the appendix. Basket weights come from the Nigeria Living Standards Survey 2018/19 [22]. We compute the food-group expenditure shares of households below the official poverty line of 137,430 naira per person per year, by geopolitical zone, and map 13 of the 19 food groups to NBS items; these cover 88% of poor households' food spending. Recipient counts are the poor population of each state from the same survey. Our computation reproduces the official 2019 poverty rates, 40.1% nationally excluding Borno and 87.7% in Sokoto. The "other grains" group (16% of spending, mainly sorghum and millet) is priced by wheat flour, the only grain product in the NBS list; results excluding it are reported as a robustness check.

Aggregation audit. Published national averages equal the unweighted mean of state prices in 78% of item–months between 2017 and 2023, and within 1% in 91%. We set ε in (U2) from this distribution. Published zonal averages are less regular. Before 2023 they cover only 10 items and lie about 25% below the mean of state prices, which points to a different item definition. In February and March 2023 they equal the mean of state prices exactly for all 43 items. From April to August 2023 they agree with the state mean at the median (ratio 0.998) but not item by item: only 8% are within 1%, and the 90th percentile of the absolute gap is 28%. The backtest therefore computes zonal averages from state prices, which isolates the value of the allocation model from errors in the published aggregates, and the live application uses the published zonal series throughout, including its August 2023 values as the base for growth.

7.2 Preliminary evidence on the cost of uniform payment

Figure 1 shows allocative efficiency (4) for three rules that need no optimisation: a uniform transfer, indexation to zonal averages, and indexation to state prices observed with a three-month delay. Need is indexed to each state's January 2019 basket cost with poor-household weights, and recipients are each state's poor population. Over February 2019 to August 2023 the uniform transfer delivered on average 82.2% of attainable protection, and 70.3% in the worst month (July 2021). Zonal indexation did worse, 78.5% on average and 60.3% at worst (August 2021), and fell below the uniform rule in 34 of 54 months. Zonal averages hide divergence within zones, so indexing to them can move money in the wrong direction. State prices with a three-month delay delivered 90.5%. These rankings hold for national weights, all-household weights, equal recipient counts and a basket without the "other grains" group. Since September 2023 a planner can use only national and zonal averages and state extremes, the information regime in which the simple zonal rule performs worst. This motivates the robust allocation of section 4.

7.3 Experimental design

All experiments are rolling and use only information available at decision time. In month t the planner sees the published zonal averages for month t and the full state panel up to month $t-1$.

Rules. We compare (i) a *flat* transfer; (ii) *zonal* indexation, $T \propto D(\bar{p}_t^Z)$; (iii) the *plug-in* rule, $T \propto \hat{D}_t$, where $\hat{D}_{st}$ is the need at zonal average prices multiplied by each state's median relative price over the previous 12 months, corrected by the median residual of the previous 12 months; (iv) the robust rule T^{rob}; and (v) the guarantee-constrained rules T^{κ} of proposition 5 with $\kappa \in \{0.5, 0.8, 0.85, 0.9, 0.95, 1\}$, all centred on the plug-in forecast.

Uncertainty sets. The price polytope uses relative-price bands with 12-month rolling centres and scales, a simultaneous conformal quantile over a 24-month calibration window, and either

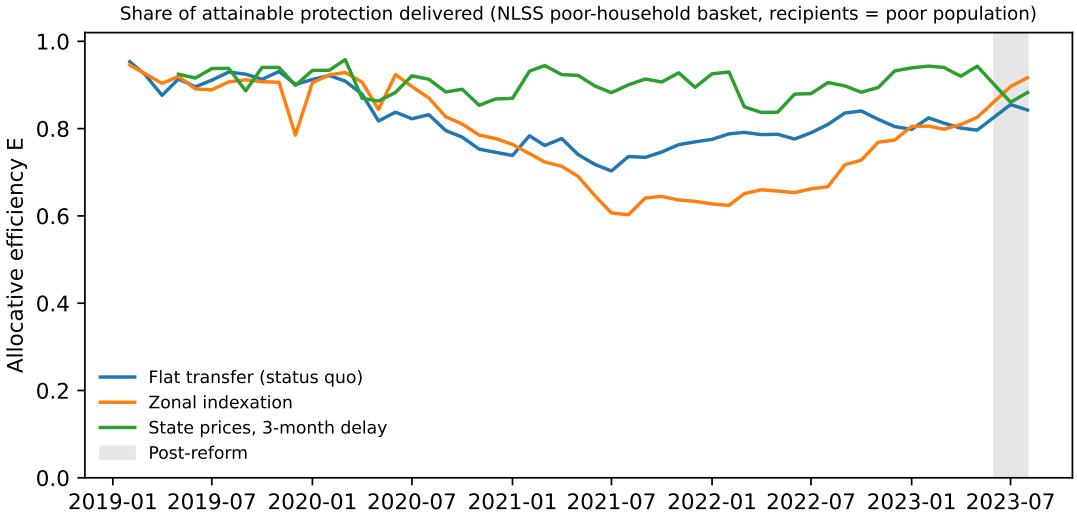


Figure 1: Allocative efficiency E of three simple rules, January 2019 base, NLSS 2018/19 poor-household weights, recipients equal to each state's poor population. Shaded: after the May 2023 reforms (genuine state data to August 2023).

zonal averages alone (I1) or zonal averages and published state extremes (I2); it is solved by proposition 3. The need box uses the same windows on the 37 need residuals (proposition 4). Both use adaptive conformal inference with target miss rate $\alpha = 0.20$ and step $\gamma = 0.02$.

Metrics. For each month we record the realised efficiency $E(T, p_t)$ of every rule at the true state prices, the guarantee G_t, the certified efficiency $\theta(T)$ of every rule (theorem 1), and whether the true needs fell inside the box. Bands need 48 months of history, so the need-box evaluation covers January 2021 to August 2023 (31 months) and the price-polytope evaluation January 2020 to August 2023 (43 months).

Robustness. Every experiment is repeated with national instead of zonal basket weights, with all-household instead of poor-household weights, and without the "other grains" group, whose mapping to NBS items is the least certain.

7.4 Results

Price polytopes are too conservative. With 1,554 price cells, a simultaneous band that covers all of them in 80% of months must be wide. Using zonal averages only (I1), the price-polytope guarantee averaged 0.28 and the robust rule realised 0.76, below the flat rule (0.80). Adding the published state extremes (I2) raised the guarantee to 0.61 and realised efficiency to 0.79. Published extremes are therefore worth a great deal, but the price-level formulation remains too cautious to beat simple rules. The need box removes this problem.

Realised versus certified efficiency. Table 2 and fig. 2 report the need-box results. The flat rule realised 0.784 of attainable protection on average and zonal indexation 0.689. The plug-in rule realised 0.922 and never fell below 0.855. It recovered 64% of the protection the flat rule loses, using only published aggregates and past state data. Its certificate, however, was weak: 0.465 on average. The guarantee-constrained rule with $\kappa = 0.9$ gave up 1.2 points of realised efficiency (0.910) and raised the certificate to 0.642. With $\kappa = 1$ the certificate reached the guarantee $G = 0.713$ but realised efficiency fell to 0.871. The certificate of every rule held in

Table 2: Realised and certified efficiency, January 2021 to August 2023 (31 months), poor-household zonal weights, need box with adaptive conformal calibration at $\alpha = 0.20$. "Holds": share of months in which realised efficiency was at least the certificate.

Rule	Realised E		Certified θ		Holds
	mean	min	mean	min	
Flat	0.784	0.703	0.403	0.241	1.00
Zonal	0.689	0.586	0.346	0.187	1.00
Plug-in	0.922	0.855	0.465	0.332	1.00
$\kappa = 0.8$	0.917	0.847	0.570	0.534	1.00
$\kappa = 0.85$	0.915	0.843	0.606	0.567	1.00
$\kappa = 0.9$	0.910	0.836	0.642	0.601	1.00
$\kappa = 0.95$	0.901	0.819	0.677	0.634	1.00
Robust ($\kappa = 1$)	0.871	0.779	0.713	0.667	1.00

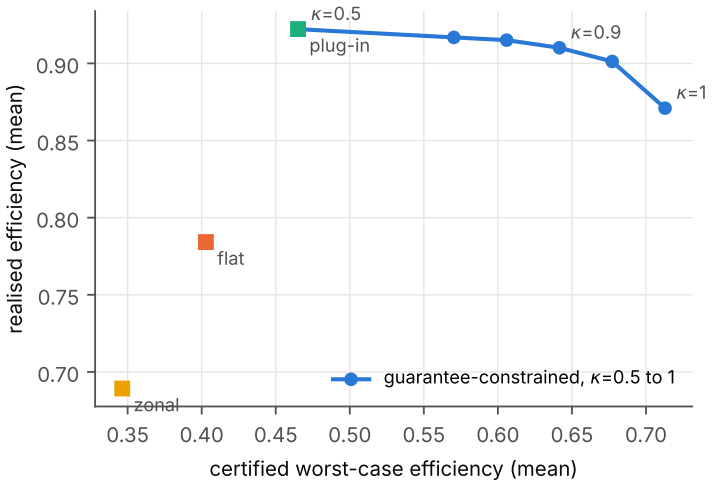


Figure 2: Frontier between certified worst-case efficiency and realised efficiency (means over 31 test months). The guarantee-constrained rules of proposition 5 dominate the flat and zonal rules on both axes.

every month, while the box covered the true needs in 87% of months against a target of 80%. Certificates can hold outside the box because they bound the worst case, which is rarely realised.

Robustness of the ranking. The ordering is identical under all four weight variants. Realised efficiency of the $\kappa = 0.9$ rule ranged from 0.907 to 0.922 and its certificate from 0.639 to 0.677; the flat rule ranged from 0.784 to 0.799 and zonal indexation from 0.678 to 0.692.

Monitoring. Figure 3 shows the mean guarantee on the 13 test months (July 2022 to August 2023) when monitored states are chosen by algorithm 2 on the 15 training months. Without delay, monitoring three states raised the guarantee from 0.721 to 0.772, five states to 0.792, ten to 0.837 and fifteen to 0.875. The first states chosen were Kano, Sokoto and FCT Abuja, followed by Jigawa, Kaduna, Katsina, Zamfara and Kebbi: populous states with many poor recipients, and states whose needs are hard to forecast from zonal averages. Greedy selection matched the exhaustive optimum for $K \leq 3$. Decision-focused selection beat accuracy-focused selection on the guarantee at every K (0.772 against 0.760 at $K = 3$, 0.837 against 0.825 at $K = 10$), while realised efficiency was similar for both (0.92 to 0.93).



Figure 3: Mean guarantee G on test months against the number of monitored states, by reporting delay and selection criterion.

Delay matters more than coverage. With a one-month delay, the guarantee stayed between 0.725 and 0.739 whatever the number of states, and with a three-month delay between 0.726 and 0.752. A late report narrows a state's band by less than the full-information band of a timely report, and the simultaneous calibration absorbs most of the gain. For a statistical agency, publishing a few states promptly is worth more than publishing many states late.

Application to September 2023 to May 2026. After August 2023 no state food prices are published. We forecast needs by (15) from August 2023 and the published zonal averages, taking both ends of the zonal growth ratio from the published series, with bands calibrated by horizon on the full 2017–2023 history. Because these bands use a longer calibration sample than the 24-month rolling windows of the backtest, their level is not directly comparable to table 2; the trend over the horizon is the object of interest. From January 2025 NBS publishes averages for 26 of our 42 items, and needs are computed on those items with weights rescaled to sum to one. Figure 4 shows the result. The guarantee fell from 0.876 in September 2023 to 0.738 by December 2024 and 0.671 by May 2026. The certificate of the $\kappa = 0.9$ rule fell from 0.835 to 0.604. The flat rule's certificate fell from 0.699 to 0.387 and that of zonal indexation from 0.786 to 0.356. Without state data, a flat payment can be guaranteed to deliver only about two fifths of attainable protection by 2026, while the guarantee-constrained rule still certifies 60%.

In May 2026 the $\kappa = 0.9$ rule paid FCT Abuja 1.56 times the flat amount, Imo 1.32 times and Ebonyi, Oyo and Cross River about 1.26 times, and paid Kogi and Plateau 0.83 and 0.80 times. Its forecast for FCT alone implies only 0.94 times. The difference is insurance: FCT's need relative to its zone's average rose by more than 60% between January 2019 and mid-2021 and fell back by 2023, so its 33-month band is the widest in the country (half-width 0.86 in logs against a median of 0.18). Because FCT holds only 2% of recipients, protecting them against that uncertainty is cheap. The premium is visible and can be reduced by monitoring FCT, which algorithm 2 selects third.

8 Discussion and policy implications

For social-protection programmes. A flat national transfer is not neutral when local prices diverge. Over 2021 to 2023 it delivered 78% of the protection the same budget could have delivered, and indexing to zonal averages, the obvious response to published data, delivered less

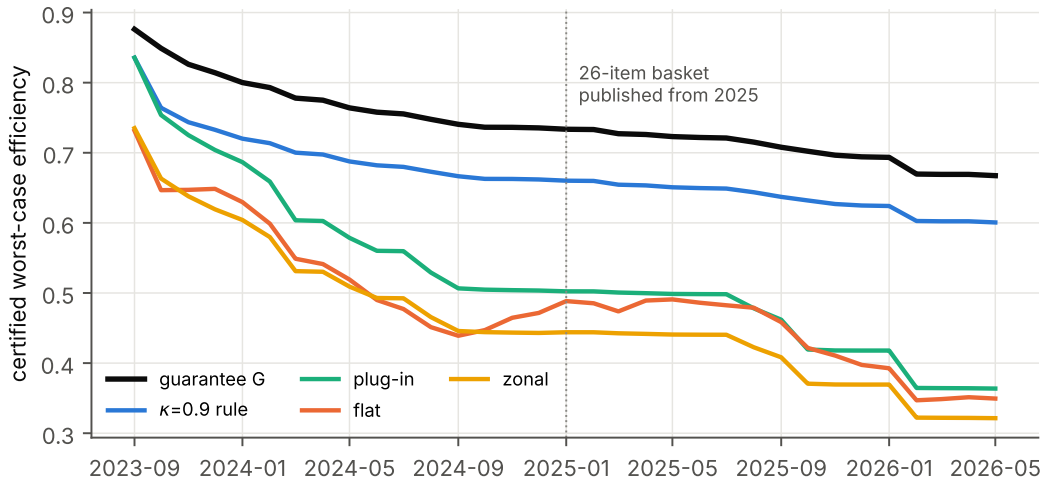


Figure 4: Guarantee and certified efficiency of four rules, September 2023 to May 2026, when only national and zonal averages and state extremes are published. Bands widen with months since the last state data (August 2023).

(69%). A programme does not need new data to do better. The guarantee-constrained rule uses only published zonal averages and the state history to August 2023, and it is a two-line formula: every state receives an insurance floor proportional to its worst-case need share, and the remaining budget is shared in proportion to forecast need. The parameter κ makes the trade-off explicit. Values near 0.9 cost about one point of realised efficiency and raise the certified worst case by about 18 points relative to the plug-in rule.

For statistical agencies. The guarantee prices information. Publishing the highest- and lowest-priced state per item, as NBS does, more than doubled the guarantee of the price-level model. Publishing a handful of populous northern states (Kano, Sokoto, Jigawa, Kaduna) together with FCT Abuja promptly would raise the guarantee more than publishing many states with a delay. Since September 2023 the absence of state tables has steadily eroded what any allocation rule can certify: by 2026 a flat payment is certified at only 39%.

Transparency. Each payment can be traced to a published average, a historical premium and a calibrated band, and each state's insurance premium is visible. The FCT example shows why this matters: a large payment relative to the forecast flags a state whose need is poorly known, which is a case for monitoring it rather than a case against the rule.

9 Conclusions, limitations and future research

We set out to answer three questions. On allocation (Q1), the robust allocation pays each state in proportion to its maximal need share, and the guarantee-constrained family of proposition 5 lets a planner keep most of the accuracy of a good forecast while insuring against its failure. On the guarantee (Q2), calibrating need boxes by adaptive conformal prediction gave certificates that held in every test month; in the backtest, with state data one month old, the guarantee averaged 0.71, and after August 2023 it fell steadily as the last state data aged, to 0.67 by May 2026. On monitoring (Q3), decision-focused greedy selection is optimal for small budgets in our data, outperforms accuracy-focused selection, and shows that reporting delay, not the number of states, limits the value of monitoring.

Limitations. The priced basket covers 88% of the food budget of poor households, and recipients are approximated by each state's poor population in 2019. The evaluation period contains a single large shock, so coverage of the conformal bands across regimes rests on the adaptive method. The staleness bands are calibrated on overlapping historical windows, and their coverage at long horizons rests on fewer independent episodes. Published averages are treated as exact for zones, and the national aggregation audit supports this only to within 1% in 91% of item–months. We ignore behavioural and price responses to the transfers themselves.

Future research. Natural extensions are joint uncertainty over prices and recipient counts, allocation at the level of local government areas with satellite or market data as side information, and monitoring designs that choose items as well as states, with explicit survey costs.

Data and code availability

Code and derived data are available at `https://github.com/Manirmaths/robust-food-indexed-transfers`[archive on Zenodo and add DOI]. Food prices are from the NBS Selected Food Price Watch [21]. NLSS 2018/19 microdata [22] are available from the World Bank Microdata Library under its terms of use and are not redistributed.

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